SRI MANAKULA VINAYAGAR ENGINEERING COLLEGE
(AN AUTONOMOUS INSTITUTION)

Department of Computer Science and Engineering

Mapping with Programme Outcomes and Programme Specific Outcomes

Applicable Program Outcomes (POs) with Justification

PO	Applicable Outcome and Justification
PO1: Engineering Knowledge	Applied machine learning principles, Python, HistGradientBoosting, physics-informed vehicle dynamics (aerodynamic drag, rolling resistance, gravitational load), and IoT telematics to build the intelligent EV fleet management prototype.
PO2: Problem Analysis	Analysed key commercial EV fleet bottlenecks, including non-linear battery degradation, static range calculation inaccuracies, peak electricity tariff expenses, and elevation-induced stranding risks.
PO3: Design/Development	Designed full-stack modular architecture including CAN-bus/IoT telematics ingestion, physics-informed range predictor, 3D elevation route optimizer with automated 150 kW DC fast-charging stops, and live operations dashboard.
PO4: Investigation	Conducted comparative benchmarks across ML algorithms (Random Forest 97.6%, Logistic Regression 98.4%, KNN 94.3%) and simulated commercial transit scenarios (585 km route) validating range assurance and 31.4% cost savings.
PO5: Engineering Tool Usage	Used Python, Scikit-learn, FastAPI/Flask, React, Vite, Leaflet.js, OpenStreetMap, PostgreSQL/SQLite, VS Code, and Git for system implementation and evaluation.
PO7: Ethics	Enforced cryptographic authentication (bcrypt hashing, JWT) and role-based access control; restricted telematics tracking strictly to vehicle operational parameters without privacy-invasive personal surveillance.
PO8: Individual and Team Work	Modules including telematics ingestion, machine learning range prediction, geospatial route optimization, and dashboard visualization were executed collaboratively across four team members.
PO9: Communication	The engineering design, mathematical formulation, and experimental outcomes are documented through comprehensive reports, architecture diagrams, PPT presentations, and live demonstration.
PO11: Life-Long Learning	Required independent exploration of physics-informed ML, battery electrochemistry, digital elevation modeling, and modern WebGIS, laying the foundation for future physical CAN-bus hardware and V2G integration.

Applicable Program Specific Outcomes (PSOs) with Justification

PSO	Applicable Outcome and Justification
PSO1: Computational Skills	Applied computer science and AI knowledge to engineer a modular, full-stack predictive EV analytics platform integrating gradient boosted decision trees and real-time geospatial routing.
PSO2: Studious Research	Converted research in EV battery state-of-health estimation and elevation-aware energy consumption into an industry-ready commercial fleet management prototype supporting green logistics.